CO2-AT3

Class Test

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1. Image Enhancement in Spatial Domain

Definition

Image Enhancement in the spatial domain improves the visual appearance of an image by directly modifying the pixel value. It increasing image quality, making important details easier to observe.

Techniques

1. Gray-level Transformations
 - Adjust brightness and contrast
 - Examples: Negative, log-transformation power law
 - Suitable for improving dark or low-contrast images.
2. Histogram Processing
 - Redistribute pixel intensity values
 - Histogram Equalization improves image contrast
 - Suitable for medical images
3. Spatial Filtering
 - Gray level transformation improves brightness
 - Histogram processing enhances contrast
 - Spatial filtering removes noise and improves image clarity

Conclusion

Spatial domain techniques are simple, fast & effective

2. Image smoothing v/s Image sharpening

Image smoothing	Image sharpening
Removes noise	Enhances edges
uses low pass filters	uses high pass filter
Produces a blurred Image	Produces a sharper image
Reduces Image details	Highlights fine details

Spatial smoothing Filters

- Replace each pixel with the average or median of neighbour pixels.
- Examples: mean filter / Median Filter
- Removes salt and pepper noise

Sharpening Filters

- Enhances intensity changes between neighbouring pixels
- Examples: - laplacian Filter
  - sobel Filter
- Improves edge visibility

Applications

Smoothing	Sharpening
- Medical Images	- Text enhancement
- Satellite Images	- object detection

Conclusion

Smoothing is used for noise removal, sharpening is used for edge enhancement

Image enhancement in frequent Domain

Definition

Frequency domain enhancement improves by modifying their frequency components using the Fourier Transform

Low-pass Filter (LPF)

- Allows low Frequency to pass
- Removes High Frequency noise
- Produces smoother image

Applications

- Noise reduction
- Medical imaging
- Image restoration

High-pass Filter

- Allows high frequency to pass
- Enhances edges and fine details
- Produces sharper images

Application

- Face recognization
- Document Processing

Comparison

Low-pass Filter	High-pass Filter
Removes noise	Enhances edges
Smooth Image	Sharp Image
Removes fine details	Highlight details

4. Image segmentation

Definition

Image segmentation divides an image into meaningful regions or objects for easier analysis

comparison

Technique	Working Principle	Application
Point detection	Detects isolated pixels	Spot detection
Line detection	Detects straight lines	Road detection
Edge Detection	Detects intensity changes	Object detection
Thresholding	Separates pixels using intensity value	Document Processing
Region based segmentation	Groups similar neighbouring pixels	Medical Imaging

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Most-suitable technique

- Edge Detection → Best for object boundary detection
- Thresholding → Best for simple images with clear background.
- Region-based segmentation → Best for complex images like MRI and CT scans

Edge

Edge Detection, Edge linking and Boundary Detection

Edge Detection

- Detects sudden change in image intensity
- Common operators:
  - Sobel
  - Prewitt
  - Canny

Edge linking

- Connects broken edge pixels to form continuous edges
- Improves object shape representation
- Uses edges continuity and connectivity.

Boundary Detection

- Identifies the complete outline of a object
- Seperates the object from the background
- Use after the edge detection and edge linking

Example

- Medical Imaging: Detects tumor boundaries
- Industrial Inspection: Find cracks in materials
- Autonomous Vehicles: Road lanes and obstacles

Conclusion

Edge detection identifies edges, edge linking connecting them, boundary detection forms complete object outline.